De Paritate et Phasi

On Parity and Phase

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An Essay on Additive Form, Numerical Invariance, and the Discipline of Applied Discovery

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"The decisive passage is not from number to number, but from an elementary identity to a disciplined instrument of inquiry."

Argumentum

THIS ESSAY begins from two supplied documents: a handwritten note entitled *Numerical Analysis*, dated August 13, 2026, and the interpretive essay *From Parity to Phase: An Applied Numerical-Analysis Interpretation*. Their common seed is the additive form

$$c = a + b,$$

its normalized consequence, and an intuition that the two summands may occupy complementary classes. The handwritten note speaks in the ancient language of odd and even; the later essay observes that ordinary parity belongs to the integers and proposes a scale-dependent phase as the rigorous continuation of that intuition.

The purpose here is neither to magnify an elementary identity into a theorem nor to dismiss it for being elementary. It is to ask a more classical question: what structure can be built when a transparent invariant is joined to a carefully defined symmetry? The answer proceeds through exact algebra, floating-point residuals, lattice phase, constrained decomposition, and standards of numerical evidence. The prospective contribution is not the sum alone. It is the architecture that makes additive closure and structural complementarity separately measurable, falsifiable, and useful.

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KEYWORDS. Numerical analysis; additive invariants; parity; phase; residuals; conditioning; constrained decomposition; multiscale structure.

Contents

Essay I

Proem: The Dignity of a Simple Form

The history of mathematics repeatedly shows that a simple form may possess two very different kinds of simplicity. It may be simple because nothing more can be said of it; or it may be simple because it is the visible edge of a deeper organization. The identity $c = a + b$ does not announce which kind it is. Taken alone, it is merely an additive statement. Taken as the exact reconstruction law of a decomposition, however, it becomes a standard against which approximation, perturbation, and computation may be judged.

The supplied handwritten note begins precisely at this threshold. It places a and b under a common total c, assigns them opposed odd–even descriptions, and searches for a normalized expression equal to unity. The accompanying applied essay performs an indispensable act of mathematical discipline: it preserves the intuition while correcting the domain. Real values such as 1.5 and 1.6 are not odd or even in the classical integer sense. If their opposition is to survive, it must be defined through something else—a lattice index, a resolution, a reflection, or a phase.

This is the proper beginning of inquiry. Mathematics advances not only by proving consequences, but by distinguishing what is given from what is proposed. In the present case, three layers must remain separate:

i. the exact algebraic layer, in which $c = a + b$ and its legitimate rearrangements hold;

ii. the numerical layer, in which finite precision and noisy observations turn exact identities into residual tests;

iii. the structural layer, in which a chosen notion of complementarity supplies information not contained in the sum itself.

Confusion among these layers would make the theory appear stronger than it is. Their separation makes it capable of becoming strong. The additive law supplies conservation; the residual supplies diagnosis; the phase rule supplies selectivity. Together they define a research program whose claims can be proved, computed, and compared.

Principle 1.1 (The classical order of construction). *State the exact form first, define the new structure second, and claim novelty only where the new structure yields a demonstrable consequence.*

The title *De Paritate et Phasi* names this passage. *Paritas* recalls parity as a discrete distinction inherited from arithmetic; *phasis* names the relation of a value to a cycle, scale, or symmetry. The argument is therefore not an abandonment of parity, but its disciplined translation from intrinsic integer property to scale-dependent relation.

Essay II

On Additive Form and the Unit Invariant

Let $a, b, c \in \mathbb{R}$ and suppose

$$c = a + b. \tag{2.1}$$

If $a \neq 0$, then

$$\frac{c-b}{a} = 1. \tag{2.2}$$

The relation is exact because the numerator equals a. Its value is not that it proves something unexpected, but that it converts reconstruction into a dimensionless checkpoint. A model that asserts (2.1) may be evaluated by its departure from (2.2).

Squaring requires the full binomial expansion:

$$\left(\frac{c-b}{a}\right)^2 = \frac{c^2 - 2bc + b^2}{a^2} = 1. \tag{2.3}$$

The cross term $-2bc/a^2$ is essential. Any expression that loses the factor c, changes the denominator, or reverses a sign ceases to be an algebraic consequence of the starting form. This correction matters beyond bookkeeping. A numerical invariant is trustworthy only if its symbolic derivation is exact.

2.1 *Verification of the handwritten derivation*

The normalized formulas recorded in the handwritten note are valid when their denominators are nonzero. From $c = a + b$, one obtains

$$\frac{a}{b} = \frac{c}{b} - 1, \qquad b \neq 0, \tag{2.4}$$

$$\frac{c-a}{b} = 1, \qquad b \neq 0, \tag{2.5}$$

$$\frac{c-b}{a} = 1, \qquad a \neq 0. \tag{2.6}$$

Thus the central unit invariant in the derivation is correct. Squaring the last identity is also correct, but its expansion must retain both the factor c in the cross term and the common denominator a^2:

$$\boxed{\left(\frac{c-b}{a}\right)^2 = \frac{c^2}{a^2} - \frac{2bc}{a^2} + \frac{b^2}{a^2} = 1}, \qquad a \neq 0. \tag{2.7}$$

Accordingly, a middle term of $-2b/a$, or a final term of b^2/a, would not be dimensionally or algebraically equivalent to the square. The corrected expression is precisely $(c-b)^2/a^2$.

One further qualification is required when reasoning backward from the squared equation. The statement

$$\left(\frac{c-b}{a}\right)^2 = 1$$

implies

$$\frac{c-b}{a} = \pm 1, \qquad \text{hence} \qquad c = b \pm a. \tag{2.8}$$

The original unsquared relation $(c-b)/a = 1$ selects the positive branch $c = a+b$. Taking a principal square root instead gives $|c-b|/|a| = 1$; it does not, without a sign condition, recover the positive branch by itself.

The numerical example verifies the complete chain exactly when the terminating decimals are regarded as rational numbers:

$$a = \frac{3}{2}, \qquad b = \frac{8}{5}, \qquad c = \frac{31}{10}.$$

Indeed,

$$\frac{a}{b} = \frac{15}{16} = \frac{c}{b} - 1, \tag{2.9}$$

$$\frac{c-a}{b} = 1, \qquad \frac{c-b}{a} = 1, \tag{2.10}$$

$$\frac{c^2 - 2bc + b^2}{a^2} = \frac{961 - 992 + 256}{225} = 1. \tag{2.11}$$

The verdict is therefore exact but qualified: the additive and normalized formulas are correct; the squared formula is correct with the full binomial expansion; and any radical reconstruction must preserve the two algebraic branches until the original sign information selects one.

2.2 The power family of unit invariants

The same forward argument is valid for every positive integer exponent. If $c = a+b$ and $a \neq 0$, then the base ratio is exactly one; consequently, for every $n \in \mathbb{N}$ with $n \geq 1$,

$$\boxed{\left(\frac{c-b}{a}\right)^n = \frac{(c-b)^n}{a^n} = \frac{1}{a^n}\sum_{k=0}^{n}\binom{n}{k} c^{n-k}(-b)^k = 1}. \tag{2.12}$$

This follows twice over: directly from $(c-b)/a = 1$, and termwise from the binomial theorem. For $n = 3$, for example,

$$\left(\frac{c-b}{a}\right)^3 = \frac{c^3 - 3c^2b + 3cb^2 - b^3}{a^3} = 1. \tag{2.13}$$

The converse depends on the parity of n. Over the real numbers, if n is odd, then

$$\left(\frac{c-b}{a}\right)^n = 1 \quad \Longrightarrow \quad \frac{c-b}{a} = 1 \quad \Longrightarrow \quad c = a+b.$$

If n is even, the same powered equation permits both signs:

$$\frac{c-b}{a} = \pm 1, \qquad c = b \pm a.$$

Thus odd powers retain the sign information of the original normalized identity, whereas even powers erase it. The case $n = 0$ is deliberately excluded: every nonzero base has zeroth power one, so it supplies no information about additive closure.

The family also has a numerical interpretation. Let

$$x = \frac{c-b}{a} = 1 + \varepsilon$$

for a small normalized defect ε. Then

$$x^n - 1 = (1+\varepsilon)^n - 1 = n\varepsilon + \binom{n}{2}\varepsilon^2 + \cdots . \tag{2.14}$$

For small ε, the powered residual is approximately $n\varepsilon$. Increasing n can therefore magnify a small failure of closure, but it also magnifies rounding error and measurement noise. The powered identities are not independent proofs of the decomposition; they are a parameterized family of diagnostics derived from the same unit invariant.

The example recorded in the sources is

$$c = 3.1, \qquad a = 1.5, \qquad b = 1.6.$$

At the displayed decimal precision, the additive relation closes: $1.5 + 1.6 = 3.1$. The normalized unit relation follows immediately. Yet on a binary floating-point machine these decimals are stored as nearby representable numbers. The computed result may therefore differ slightly from its exact mathematical value, depending on evaluation order and rounding. What was a tautological equality in exact arithmetic becomes a small experiment in representation.

This distinction is central to numerical analysis. Exact mathematics asks whether a statement follows from its assumptions. Numerical mathematics asks additionally whether the computation of that statement is stable, whether the data are known accurately, and whether the chosen diagnostic amplifies or suppresses error.

For an observed triple $(\widehat{a}, \widehat{b}, \widehat{c})$, define the direct normalized defect

$$r_a = \frac{\widehat{c} - \widehat{b}}{\widehat{a}} - 1, \qquad \widehat{a} \neq 0. \tag{2.15}$$

If $r_a = 0$, the unit relation closes in the performed arithmetic. But the interpretation of a nonzero value depends upon scale. A residual of 10^{-8} may be negligible in measured data and unacceptable in a carefully conditioned computation. Thus no residual speaks without a tolerance model.

The additive form also reveals an identifiability limit. Given only c, there are infinitely many pairs (a, b) satisfying $a + b = c$. The equality reconstructs but does not separate. Any method that claims to recover distinct components must add structure: bounds, smoothness, sparsity, phase opposition, probabilistic priors, or physical laws. The odd–even intuition in the source note is best understood as a proposed structural condition of this kind.

Principle 2.1 (Invariant and information). *An invariant tests whether a proposed decomposition closes; it does not, by itself, determine the decomposition. Identification requires an additional principle.*

Essay III

From Parity to Phase

Classical parity is defined on the integers. An integer n is even when $n = 2k$ and odd when $n = 2k+1$ for some $k \in \mathbb{Z}$. No corresponding assertion is intrinsically true of 1.5 or 1.6. To call such values odd and even without further definition would be to extend a word while withholding its rule.

There are, however, rigorous structures that preserve the intended two-class opposition. The first is *lattice parity*. Choose a scale $h > 0$ and associate a value x with an integer index

$$k_h(x) = \mathrm{round}\left(\frac{x}{h}\right). \tag{3.1}$$

The parity of $k_h(x)$ then assigns a class to x relative to resolution h. This is not parity of the real number itself; it is parity of its selected lattice representative. The distinction is precise and productive.

A continuous companion is the phase function

$$p_h(x) = \cos\left(\pi \frac{x}{h}\right). \tag{3.2}$$

At lattice points $x = kh$, one has $p_h(x) = (-1)^k$. Even indices produce $+1$, odd indices produce -1, and off-lattice values produce a graded measure between these extremes. The scale h is not an inconvenience to be hidden. It declares the resolution at which alternation is being examined.

For two components, define the soft complementarity score

$$\chi_h(a,b) = \frac{1 - p_h(a)p_h(b)}{2}. \tag{3.3}$$

At lattice points, $\chi_h = 1$ when the indices have opposite parity and $\chi_h = 0$ when they have the same parity. Between lattice points it measures degree rather than issuing a brittle verdict. Additive closure and complementarity are now separate observables: one may hold while the other fails.

The second rigorous structure is *operator parity*. For a function f defined on a reflection-symmetric domain,

$$f_{\mathrm{even}}(x) = \frac{f(x) + f(-x)}{2}, \tag{3.4}$$

$$f_{\mathrm{odd}}(x) = \frac{f(x) - f(-x)}{2}. \tag{3.5}$$

Then $f = f_{\mathrm{even}} + f_{\mathrm{odd}}$, the first component is invariant under reflection, and the second changes sign. Here parity is not attached to sampled magnitudes at all; it is attached to the action of a symmetry operator. This decomposition is classical, exact, and often more natural when the data are functions or fields.

These two continuations should not be conflated. Lattice parity concerns alternation across indexed locations or quantized values. Operator parity concerns symmetry under reflection. A successful application must choose the structure dictated by the phenomenon.

The phrase "asymmetric by one" in the handwritten note can consequently receive several testable meanings: separation by one lattice index; opposition of the signs of p_h; or unit distance after a declared normalization. Each choice yields a different theory. The act of definition converts suggestive language into an algorithm.

Principle 3.1 (Relational parity). *For real-valued data, the productive heir of parity is not an intrinsic odd–even label, but a relation among a value, a scale, and a symmetry.*

ESSAY IV

Residual, Error, and Condition

The direct residual (2.15) is natural but becomes sensitive when $|a|$ is small. Division by a small denominator magnifies perturbations. For broad diagnostic use, the supplied applied essay proposes a symmetric scaled residual:

$$\eta(a, b, c) = \frac{|c - a - b|}{\max\{1, |a| + |b| + |c|\}}. \tag{4.1}$$

This quantity is dimensionless, bounded against a vanishing single component, and symmetric in the magnitudes contributing to the scale. In exact arithmetic under (2.1), it is zero. With noisy data or rounded computation, it expresses the defect relative to the size of the triple.

Suppose the exact variables satisfy $c = a + b$, while observed values are $\widehat{a} = a + \delta a$, $\widehat{b} = b + \delta b$, and $\widehat{c} = c + \delta c$. The unscaled closure defect is

$$\widehat{c} - \widehat{a} - \widehat{b} = \delta c - \delta a - \delta b. \tag{4.2}$$

Hence

$$|\widehat{c} - \widehat{a} - \widehat{b}| \leq |\delta a| + |\delta b| + |\delta c|. \tag{4.3}$$

This elementary inequality already gives a falsifiable expectation: if error bounds on the three measurements are known, a defect greatly exceeding their sum signals model failure, corruption, or underestimated uncertainty.

The phase score has a different sensitivity. Differentiating (3.2) gives

$$p_h'(x) = -\frac{\pi}{h} \sin\left(\pi \frac{x}{h}\right). \tag{4.4}$$

Thus its local sensitivity scales like $1/h$. Fine resolutions distinguish closely spaced structure but also magnify perturbations in the phase argument. A scale selected solely for visual agreement may therefore produce unstable classifications. The scale must be justified by sampling density, physical periodicity, cross-validation, or a multiscale criterion.

It is useful to regard $(\eta(a, b, c), \chi_h)$ as a two-coordinate diagnostic. The first coordinate asks whether the sum closes. The second asks whether the components occupy opposed phase classes. Four qualitative regimes follow:

Closure	Complementarity	Interpretation
good	strong	structurally consistent decomposition
good	weak	reconstruction succeeds; class model fails
poor	strong	phase persists; amplitude or conservation fails
poor	weak	decomposition and structural model both fail

Essay V

The Architecture of a Method

The idea becomes algorithmic when decomposition is posed as constrained optimization. Given an observation c, seek components a and b by minimizing

$$J(a,b) = |c - a - b|^2 + \lambda\,\Phi_h(a,b) + \alpha R_a(a) + \beta R_b(b), \tag{5.1}$$

where Φ_h penalizes failure of phase complementarity, R_a and R_b encode properties such as smoothness or sparsity, and $\lambda, \alpha, \beta \geq 0$ regulate the tradeoffs. One possible choice is $\Phi_h = (1 - \chi_h)^2$, although other penalties may have better optimization properties.

Equation (5.1) makes explicit what the additive identity cannot do alone. The data term enforces reconstruction. The phase term favors opposed classes. The regularizers express domain knowledge. The method can now be tested for existence of minimizers, sensitivity to parameter choice, convergence of its numerical solver, and performance against baselines.

Four application families appear especially natural.

5.1 Alternating artifacts in signals and images

Grid-scale oscillation often has a checkerboard or alternating signature. A phase-aware component may isolate this pattern while a smoother component retains the underlying signal. Unlike a generic high-pass filter, the proposed method can enforce exact or near-exact reconstruction and can target a declared alternation scale. Its merit would need to be measured against Fourier, wavelet, total-variation, and learned denoising methods.

5.2 Complementary updates in numerical solvers

Red–black relaxation divides a regular grid into alternating node sets so that one class can be updated from the other. A generalized continuous phase may help extend this logic to irregular meshes or weighted graphs. The relevant outputs would be convergence rate, parallel efficiency, communication cost, and robustness to mesh distortion. If the phase assignment merely reproduces graph coloring at greater cost, the result is exposition rather than breakthrough. If it yields stable near-bipartite scheduling where strict coloring is awkward, it becomes interesting.

5.3 Two-source inverse problems

When c is a mixture of latent sources, additive closure alone leaves the pair non-identifiable. A phase prior can contribute distinguishing information when the physics predicts alternation in time, space, scale, or state. The scientific obligation is to show that the prior corresponds to the mechanism rather than merely forcing a convenient answer. Synthetic recovery tests should precede real-data claims, and uncertainty in the recovered sources should be reported.

5.4 *Fault and anomaly detection*

In a calibrated system, closure and complementarity may fail for different reasons. A rise in $\eta(a, b, c)$ can indicate conservation loss, sensor drift, or corrupted amplitude. A fall in χ_h can indicate synchronization where opposition was expected. Monitoring the pair may distinguish structural faults from magnitude faults. Here interpretability is an advantage: each alarm has a mathematical meaning.

Across all four families, the scale parameter h makes a multiscale analysis possible. The same observation can be examined over a sequence $h_1 > h_2 > \cdots > h_m$. Persistent complementarity across scales suggests robust structure; a sharp transition may reveal a characteristic resolution. Such a scale portrait resembles the spirit of multiresolution analysis without being identical to it. Comparison, not analogy alone, must determine whether it adds value.

Principle 5.1 (The location of novelty). *The novelty cannot reside in $c = a + b$. It must reside in a well-defined phase structure, a stable recovery or diagnostic algorithm, and an application in which that structure yields a reproducible advantage.*

Essay VI

The Measure of an Applied Breakthrough

The language of breakthrough should be the conclusion of a program, not its premise. A credible program proceeds by increasingly severe trials.

First comes *symbolic integrity*. Every algebraic transformation must be checked, every denominator restriction stated, and every new term defined. The corrected square in (2.3) is exemplary: a missing factor would contaminate all later numerical conclusions.

Second comes *finite-precision analysis*. Decimal, binary64, and arbitrary-precision computations should be compared. Residuals should be interpreted through rounding-error bounds rather than through visual smallness. The behavior near $a = 0$ should be studied separately because the direct normalization changes condition there.

Third comes *perturbation*. Controlled noise should be added to a, b, and c. Inequality (4.3) supplies a first bound; more refined probabilistic analysis may describe typical rather than worst-case behavior. Phase stability should be mapped as a function of h and of distance from class boundaries.

Fourth comes *ablation*. The additive term, phase term, and regularizers in (5.1) should be removed one at a time. If the claimed gain survives when the phase term is removed, phase is not the source of the gain. If it disappears when ordinary smoothness regularization is removed, the method may be a repackaging of a known prior.

Fifth comes *baseline comparison*. The proposal must meet established methods on their strongest reasonable settings. Depending on the application, these may include least squares, Fourier and wavelet filtering, graph coloring, alternating-direction schemes, total variation, or domain-specific estimators. A result is meaningful when the dataset, metric, parameter-selection rule, and computational budget are disclosed.

Finally comes *replication*. Code, data provenance, random seeds, and tolerances should be recorded. A held-out dataset should test whether performance generalizes beyond the example that inspired the formulation. Claims of stability require repeated trials; claims of convergence require either proof under stated assumptions or systematic numerical evidence that clearly stops short of proof.

The hierarchy of possible outcomes is itself useful:

1. *Interpretive success*: the formulation becomes mathematically coherent.

2. *Algorithmic success*: a stable method implements the coherent formulation.

3. *Applied success*: the method improves a defined task under fair comparison.

4. *General success*: the advantage persists across datasets, scales, and independent tests.

Only the latter stages warrant the strongest language. Yet the earlier stages are not failures. Turning an ambiguous intuition into a precise, testable object is itself legitimate mathematical work. The classical virtues are proportion, clarity, and proof appropriate to the claim.

The first experiment should therefore be narrow. Choose one phenomenon with genuine alternating structure; select h by a declared rule; compute both $\eta(a, b, c)$ and χ_h; solve the constrained decomposition; compare against a strong baseline; and publish the cases in which the method fails. A small honest experiment has more scientific force than an expansive claim without a decisive test.

Essay VII

Epilogue: From Seed to Instrument

The two source documents together enact a fruitful movement of thought. The handwritten note offers a compact seed: a whole divided into two parts, a desired opposition between those parts, and a normalization toward unity. The applied essay subjects that seed to numerical discipline: it corrects the algebra, respects the domain of parity, and proposes phase as a scale-aware continuation.

The resulting framework has a clear logical order. The additive identity

$$c = a + b$$

supplies reconstruction. The residual $\eta(a, b, c)$ measures failure of closure. The phase function p_h and complementarity score χ_h measure a separate structural hypothesis. A constrained objective joins these elements without confusing them. Perturbation analysis, conditioning, benchmarks, and ablation then determine whether the construction is useful.

This is why elementary beginnings should be treated with neither credulity nor contempt. Their proper test is architectural. Can the primitive form support definitions that are rigorous? Can those definitions support computations that are stable? Can those computations expose something that established methods do not expose as clearly, cheaply, or reliably?

If the answer is yes for a well-chosen application, the advance will not consist in discovering that a sum may be rearranged. It will consist in discovering that additive closure and phase complementarity, measured independently and enforced jointly, reveal a structure worth computing. If the answer is no, the inquiry will still have produced a clean account of where the intuition ends.

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The most promising next act is therefore modest and exact: define one phase rule, derive one perturbation bound, implement one reproducible solver, and test one real alternating phenomenon. Through such limits an idea acquires form. Through such form it becomes an instrument.

Sources and Selected Background

Supplied documents

1. Ivankovic, Jasmin. *Numerical Analysis*. Handwritten one-page note, August 13, 2026.
2. Ivankovic, Jasmin. *From Parity to Phase: An Applied Numerical-Analysis Interpretation*. Seven-page essay, August 2026.

The exact additive formulation and the motivating numerical example originate in the handwritten note. The correction of the squared expansion, the scale-dependent lattice phase, the symmetric residual, the complementarity score, and the staged validation program are developed in the supplied applied essay. The present essay synthesizes those materials into a classical argumentative form and extends their methodological organization. No claim is made that generalized parity, convergence, identifiability, or applied superiority has already been proved.

Selected mathematical background

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3. Trefethen, Lloyd N., and David Bau III. *Numerical Linear Algebra*. Philadelphia: SIAM, 1997.

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